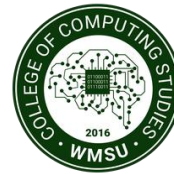




Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City



CAPSTONE PROJECT AND RESEARCH CAPSULE PROPOSAL

Proposed Research Title

VisionPool: A Real-Time Pool Rule Compliance Monitoring System Using Object Detection and Automated Flagging for Administrative Oversight

Name of the Proponent(s)

Solis, Luisa Mae C.

Research Discipline/Topic

Computer Vision & AI-Powered Safety Systems

Sector/Priority Areas/SDG

Sector: Public Safety & Security

SDG: SDG 3(Good health and Well Being), SDG 11(Sustainable Cities and Communities), SDG 9(Industry, Innovation, and Infrastructure)

Rationale/Introduction

Public swimming pools are recreational spaces that require strict adherence to safety and operational rules to ensure the well-being of patrons and the efficiency of pool management. Common pool policies include the use of wristbands to verify paid entry, mandatory proper swimming attire, and restrictions on jumping or diving in designated areas. However, manual enforcement of these rules is often inconsistent, labor-intensive, and prone to human error, particularly during peak hours when pool areas experience high foot traffic.

The proliferation of affordable IP cameras and advances in deep learning-based object detection have opened opportunities for automated visual monitoring systems in public spaces. YOLOv8 (You Only Look Once version 8), one of the most efficient real-time object detection frameworks available today, enables accurate detection of persons and objects in live video streams with minimal computational overhead. By leveraging such technology, pool facilities can automate the detection of rule violations and instantly notify management personnel.

VisionPool is a proposed real-time pool rule compliance monitoring system that uses object detection to automatically identify violations such as missing wristbands, improper swimming attire, and prohibited jumping or diving activities. The system captures screenshots of flagged incidents and transmits them to a web-based administrative dashboard, providing pool staff with an efficient, data-driven tool for rule enforcement and incident documentation.

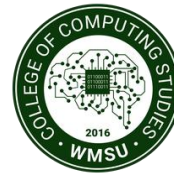
Objectives

General Objective:

To design, develop, and evaluate a real-time pool rule compliance monitoring system that uses object detection to automatically flag violations and relay them to an administrative dashboard for oversight and documentation.

Specific Objectives:

- To develop a YOLOv8-based detection model capable of identifying wristbands, swimming attire compliance, and jumping/diving activities from live camera feeds.
- To implement an automated flagging and screenshot capture mechanism that records violation incidents with timestamps and violation type labels.
- To develop a web-based administrative dashboard using Django that displays flagged incidents in real time and maintains a searchable violation log.
- To integrate the detection pipeline with an IP camera or webcam input via OpenCV for real-time processing.



- To evaluate the accuracy, precision, and recall of the detection model on a locally collected dataset of pool environment images.

Features of the Proposed Study

- Design and development of a YOLOv8-based object detection model trained to identify (a) wristbands on pool patrons, (b) appropriate vs. inappropriate swimming attire, and (c) jumping or diving actions in designated no-jump zones.
- Real-time video processing from a single fixed camera (IP camera or webcam) using OpenCV.
- Automatic screenshot capture upon detection of a violation, tagged with timestamp and violation type.
- Development of a Django-based administrative web dashboard for displaying, filtering, and reviewing flagged violation incidents.
- Local deployment within a pool facility environment for testing and evaluation purposes.

Methodology

The study will follow an iterative development methodology consisting of the following phases:

Phase 1: Data Collection and Annotation

Images and video clips will be collected from pool environments depicting patrons with and without wristbands, wearing proper and improper swimming attire, and performing jumping or diving activities. Images will be manually annotated using Roboflow or Labellmg to generate bounding box labels for each detection class. Supplementary datasets from Roboflow Universe will be explored to augment the training set.

Phase 2: Model Training and Validation

A YOLOv8 model (nano or small variant) will be fine-tuned on the annotated dataset using the Ultralytics training pipeline in Python. The dataset will be split 80/10/10 for training, validation, and testing. Model performance will be evaluated using standard metrics including mean Average Precision (mAP), precision, recall, and F1-score.

Phase 3: System Integration

The trained model will be integrated with an OpenCV video capture pipeline for real-time frame processing. A violation detection module will apply threshold-based rules to trigger flagging events. Upon flagging, the system will save a screenshot to the server and transmit incident data (timestamp, violation type, image path) to the Django backend via a local API call.

Phase 4: Administrative Dashboard Development

A Django web application will serve as the admin interface. The dashboard will display a real-time violation feed, support filtering by date and violation type, and allow administrators to review and annotate incident records. SQLite will be used as the database backend for the prototype.

Phase 5: Testing and Evaluation

The complete system will be tested in a simulated pool environment to assess end-to-end performance including detection accuracy, flagging latency, and dashboard responsiveness. System usability will be evaluated using a structured questionnaire administered to pool staff evaluators.

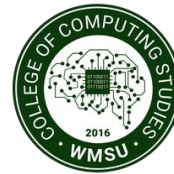
Expected Output

The VisionPool research/project is expected to produce the following outputs:

1. Product – A functional prototype system consisting of (a) a fine-tuned YOLOv8 object detection model achieving a target mean Average Precision (mAP) of at least 80% for detecting wristband presence, swimming attire compliance, and jumping/diving actions; (b) a real-time monitoring application integrating OpenCV-based video processing with automated screenshot capture and incident flagging; and (c) a Django-based



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administrative web dashboard for reviewing, filtering, and managing violation logs. This prototype can be deployed for pilot testing in at least one local pool facility and may serve as a foundation for a commercial pool-monitoring solution for resorts, fitness centers, and recreational facilities.

2. Publication – A capstone research paper documenting the system design, detection model performance, and usability evaluation results, contributing to the body of knowledge on applied computer vision for public safety and facility management. This may be submitted to a local or regional IT/computing research conference or student research journal.

3. Places and Partnerships – A locally collected and annotated image dataset (target: at least 1,000–1,500 labeled images across the three violation categories) that can serve as a reusable resource for future capstone or thesis projects involving object detection in recreational or public-space monitoring contexts. The project may also establish a pilot partnership with a local pool facility (e.g., a resort or sports complex in Zamboanga City) for system testing and data collection.

4. People Services – Development of the proponent's technical competencies in deep learning model training, computer vision pipeline development, and full-stack web application development (Django), contributing to the growing pool of AI/ML-skilled IT graduates in the region.

5. Policies – Documented recommendations for pool facility operators on integrating automated compliance monitoring into their standard operating procedures, which can inform internal facility policies on safety enforcement and incident documentation.

Target Beneficiaries

•**Pool Administrators and Security Staff** – VisionPool reduces the burden of manual monitoring by providing automated, real-time detection of violations. The admin dashboard gives staff a centralized view of flagged incidents, enabling faster and more consistent enforcement decisions.

•**Pool Patrons** – A more consistently enforced pool environment improves safety and fairness for all users, as rule compliance is monitored objectively by the system rather than subjectively by individual staff.

•**Pool Facility Owners and Operators** – Automated monitoring reduces staffing costs associated with manual oversight. Violation logs provide data for management review, policy adjustment, and liability documentation.

•**Researchers and Developers in the CV/AI Field** – VisionPool contributes a localized dataset and implementation framework for object detection applied to recreational facility monitoring, which may serve as a reference for future related studies.

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